

CS & IT ENGINEERING



THEORY OF COMPUTATION

Pushdown Automata

Lecture – 02



By– Venkat sir



Recap of Previous Lecture



Topic

Pushdown Automata

?????

Pushdown Automata Construction

$$L = \{a^n b^n \mid n \geq 1\}$$



Topics to be Covered



Topic

Push down automata Construction

Topic

??

DPDA & NPDA

Topic

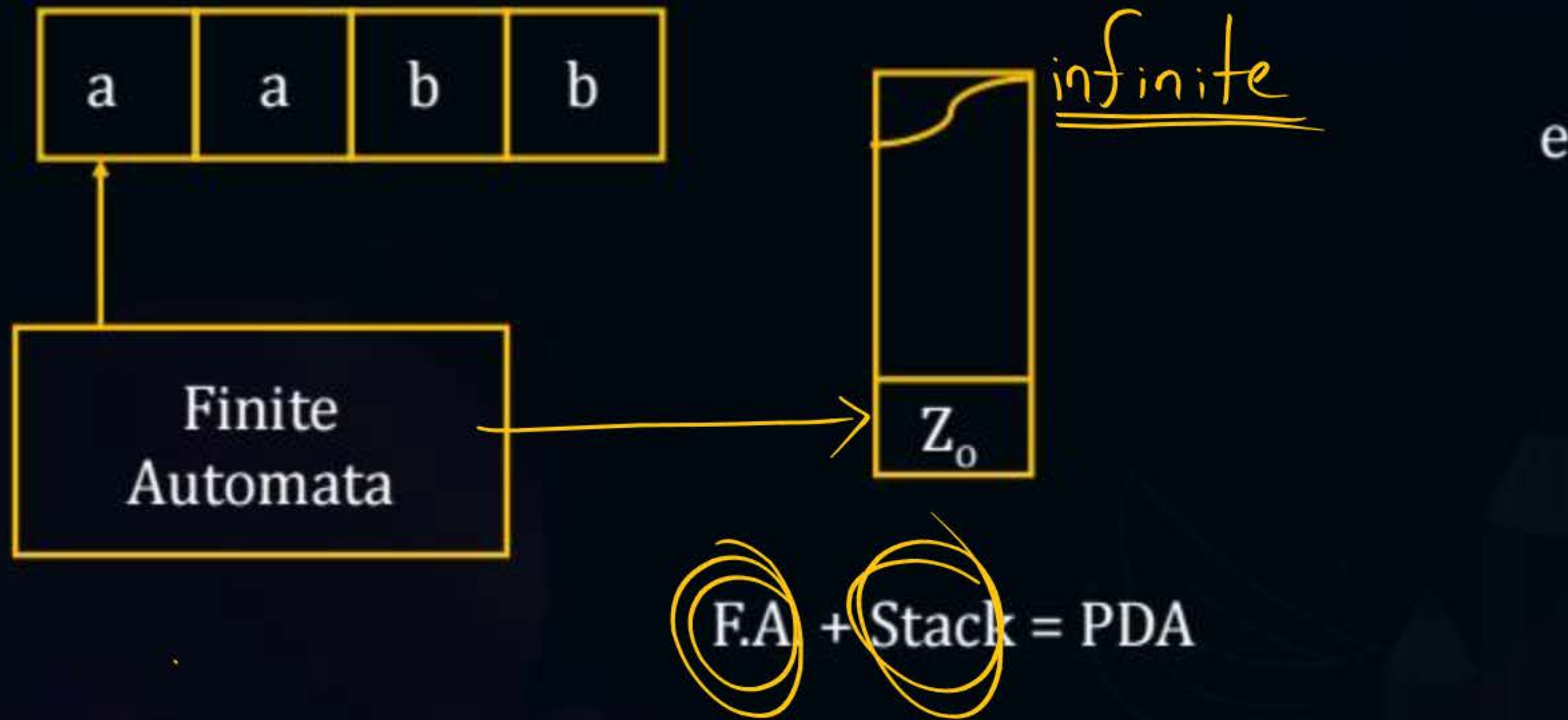
??

CF & Derivations

Topic

??

Pushdown Automata





Topic : PDA



- ① Finite Automata having additional power form of stack known as Push down automata.
- ② Size of stack in Push Down automata is infinite
- ③ There exist only one type of push down automata i.e. “language recognisor”
- ④ Push down automata can accept language in deterministic way (or) non-deterministic way

Formal Definition

PDA $(Q, \Sigma, \delta, q_0, F, Z_0, \Gamma)$

Q :- Finite number of states

Σ :- Input alphabet

q_0 :- initial state

F :- set of final states

Z_0 :- initial stack symbol

Γ :- stack alphabet

δ :- transition function

$$Q \times \Sigma \cup \{\epsilon\} \times \Gamma \rightarrow Q \times \Gamma$$



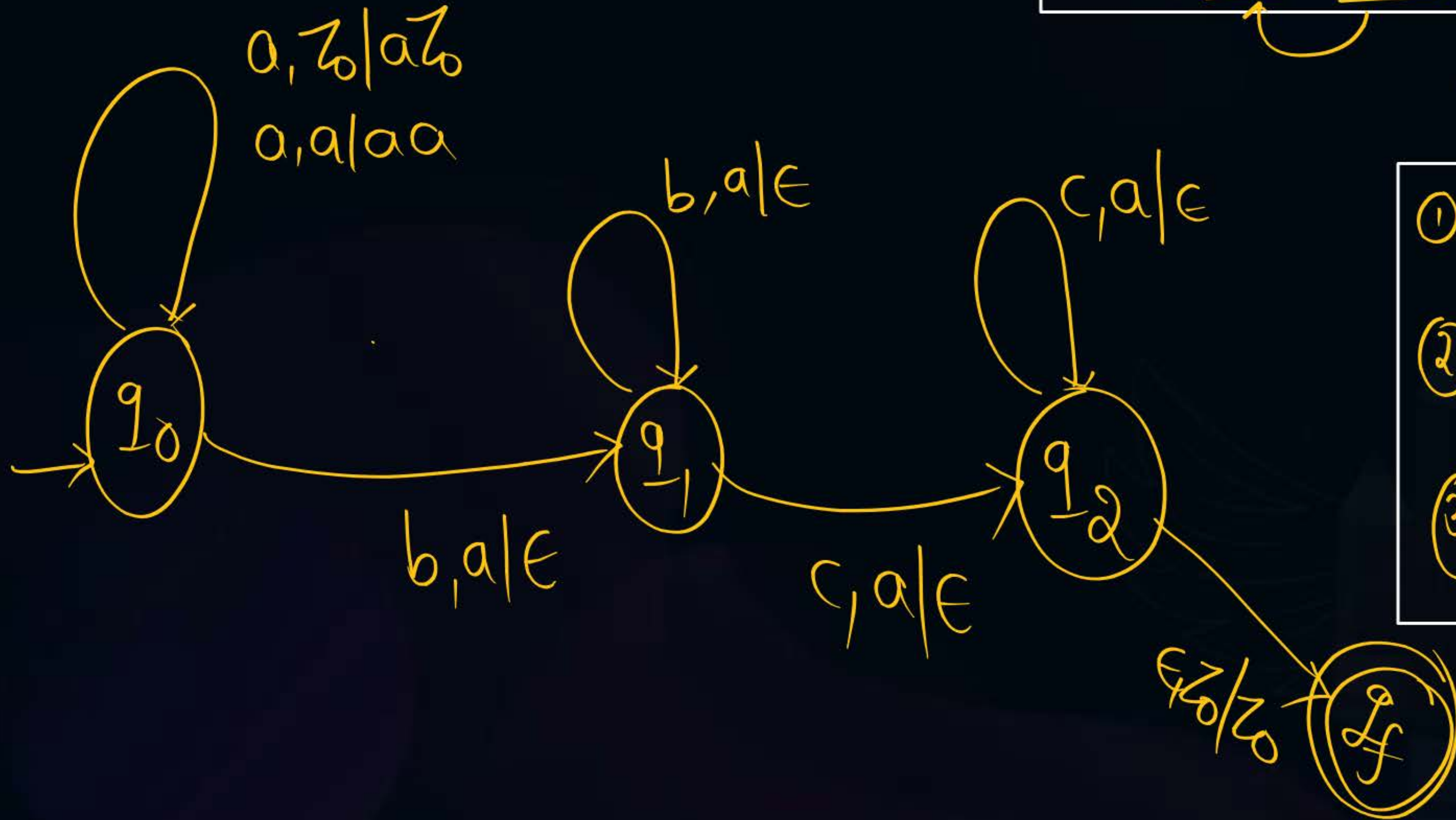
Topic : Pushdown Automata

Construct PDA for

$$L = \{ \underline{a^{n+m}} \underline{b^n} \underline{c^m} \mid n, m \geq 1 \}$$

Logic

- ① a's $\rightarrow$ Push
- ② b's $\rightarrow$ pop a's
- ③ c's $\rightarrow$ pop a's





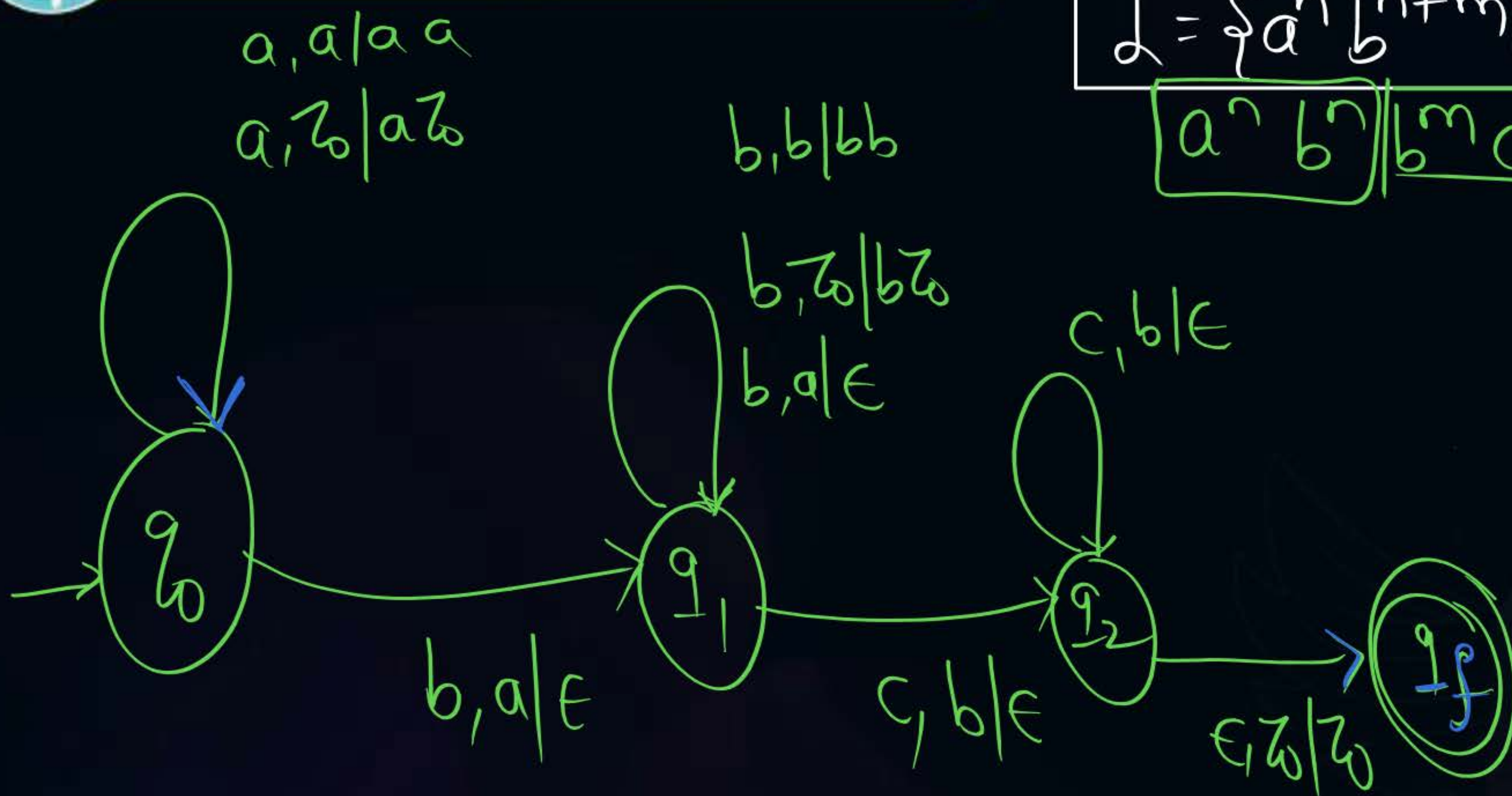
Topic : Pushdown Automata

Construct PDA for

$$L = \{a^n b^{n+m} c^m \mid n, m \geq 1\}$$

$$\boxed{a^n b^n} \mid \boxed{b^m c^m}$$

Logic



① a 's $\rightarrow$ Push into stack

② b 's $\rightarrow$ pop a 's
 b 's $\rightarrow$ z_0 ? push
 $b, b \rightarrow$ push

③ c 's $\rightarrow$ pop b 's



Topic : Pushdown Automata



$a, \epsilon / a\epsilon$ ✓✓

$b, \epsilon / b\epsilon$ ✓✓

$a, a / aa$

$b, b / bb$

$a, b / \epsilon$

$b, a / \epsilon$

$\epsilon, \epsilon / \epsilon$

(empty stack)

Construct PDA for [empty stack]

$L = \{ x \mid n_a(x) = n_b(x) \}$ $x \in (a+b)^*$

$\{ \epsilon, ab, ba, \dots \}$ Logic:-

① $(a, b) \rightarrow pop$

$(b, a) \rightarrow pop$

② Remaining all path



Topic : Pushdown Automata



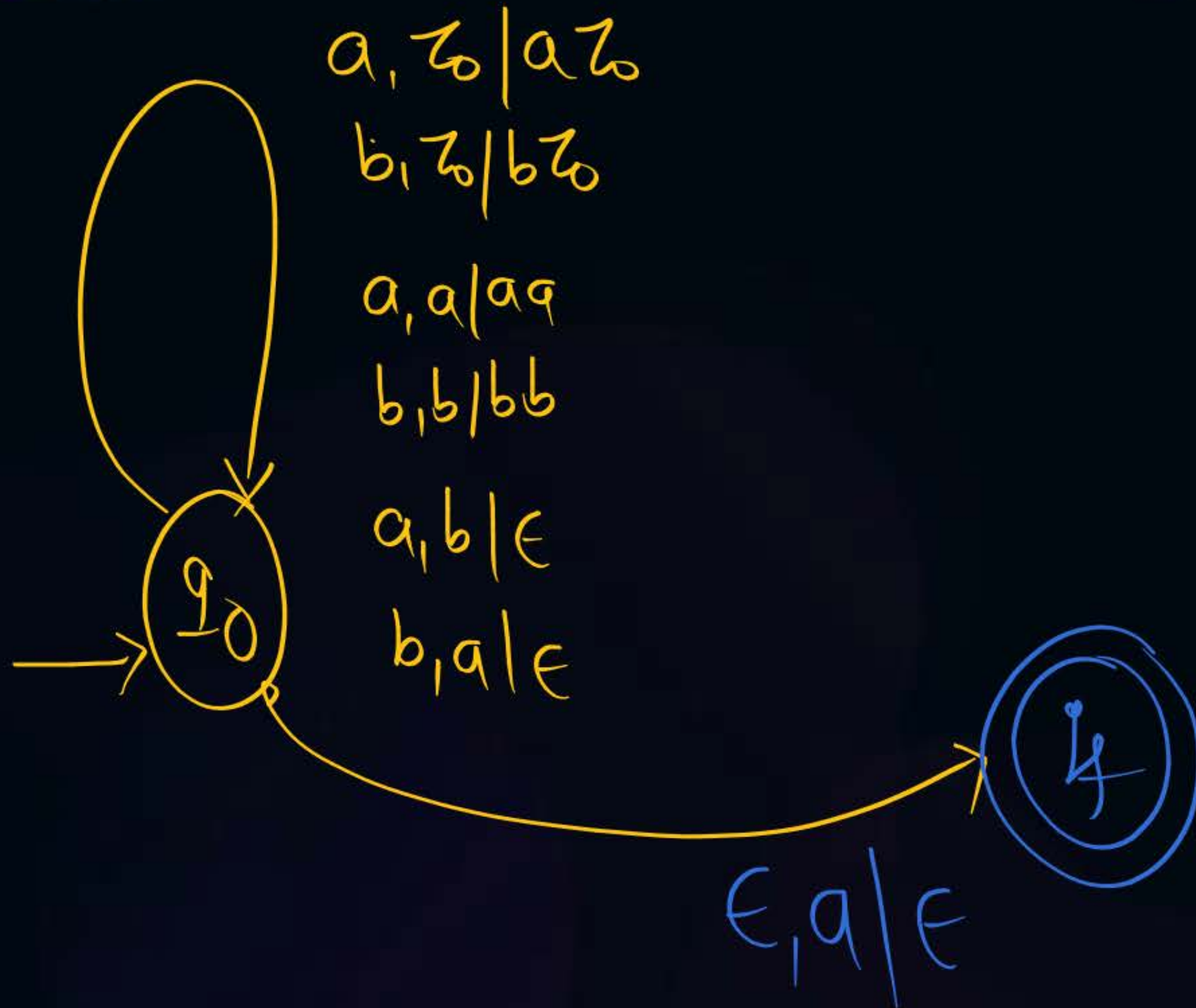
Construct PDA for

$$L = \{x \mid n_a(x) > n_b(x)\} \quad \underline{x \in (a+b)^*}$$

Logic:

① $n_a(x) = n_b(x)$

② $(\epsilon, a) \rightarrow \text{accept}$



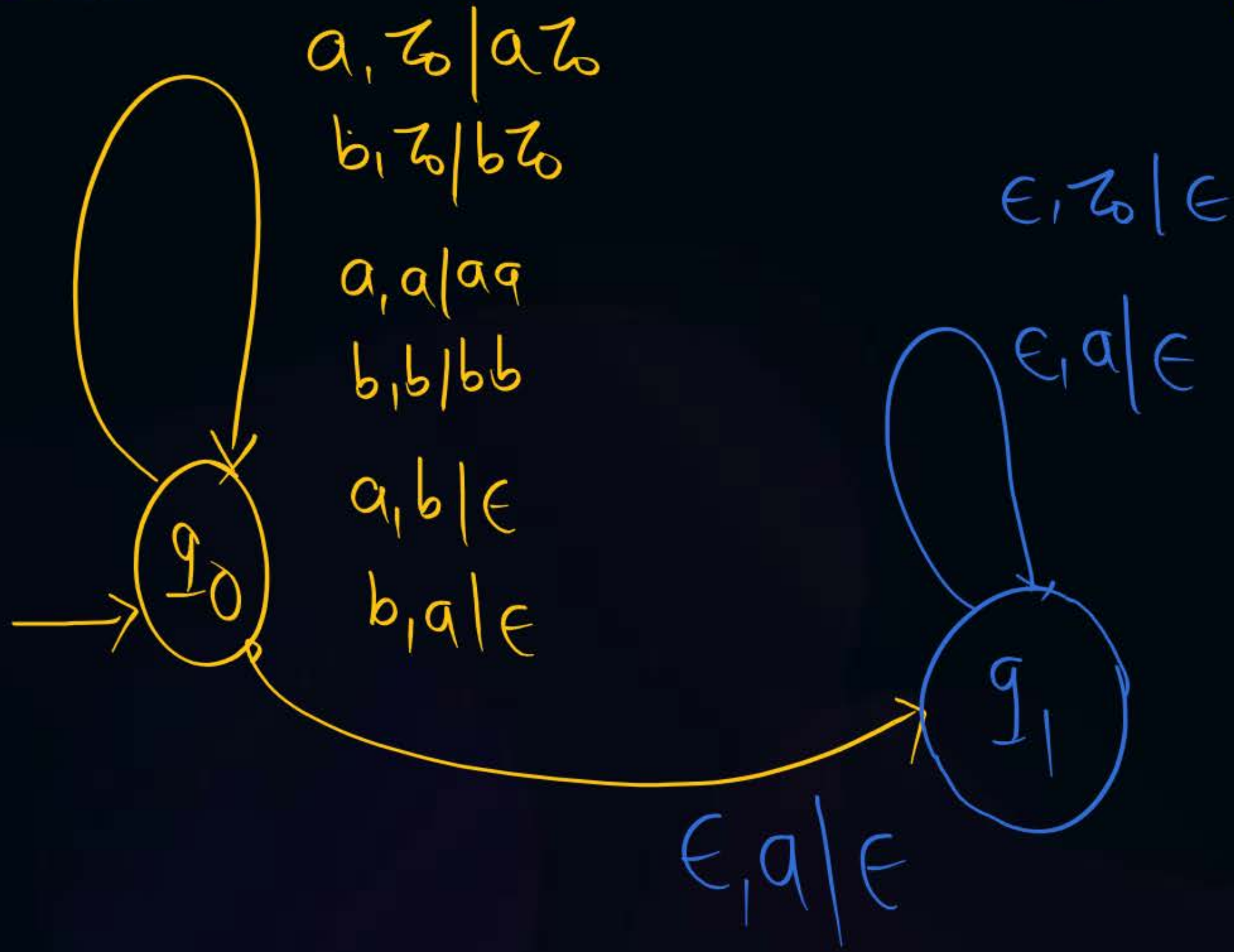
Final state



Topic : Pushdown Automata

Construct PDA for

$$L = \{x \mid n_a(x) > n_b(x)\} \quad \underline{x \in (a+b)^*}$$



Logic:

① $n_a(x) = n_b(x)$

② $(\epsilon, a) \rightarrow \text{accept}$

empty stack



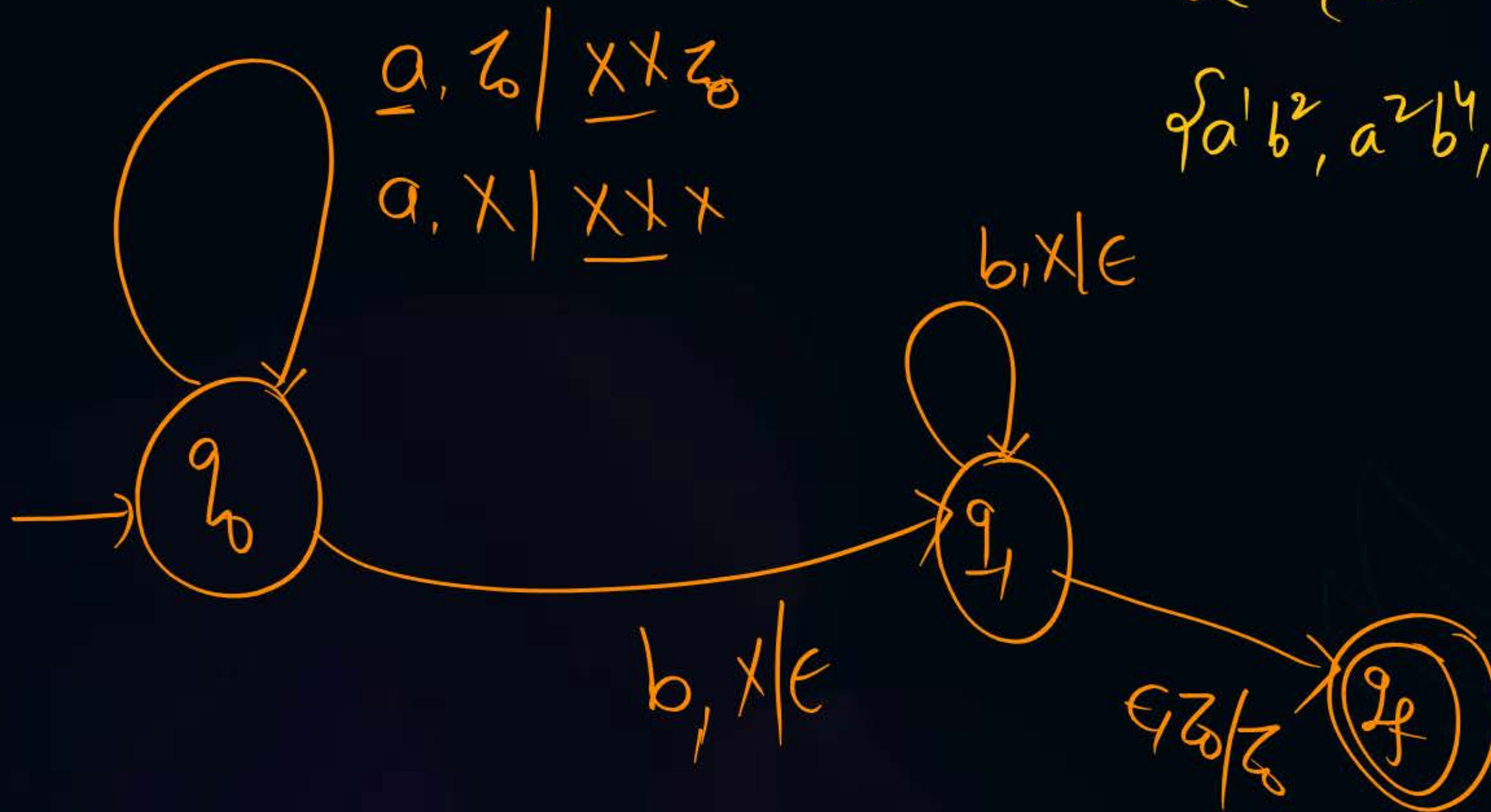
Topic : Pushdown Automata



Construct PDA for

$$L = \{a^n b^{2n} / n \geq 1\}$$

$\{a^1 b^2, a^2 b^4, a^3 b^6, \dots\}$ Logic



① 1 a's $\rightarrow$ push XX

② 1 b's $\rightarrow$ pop 1 X

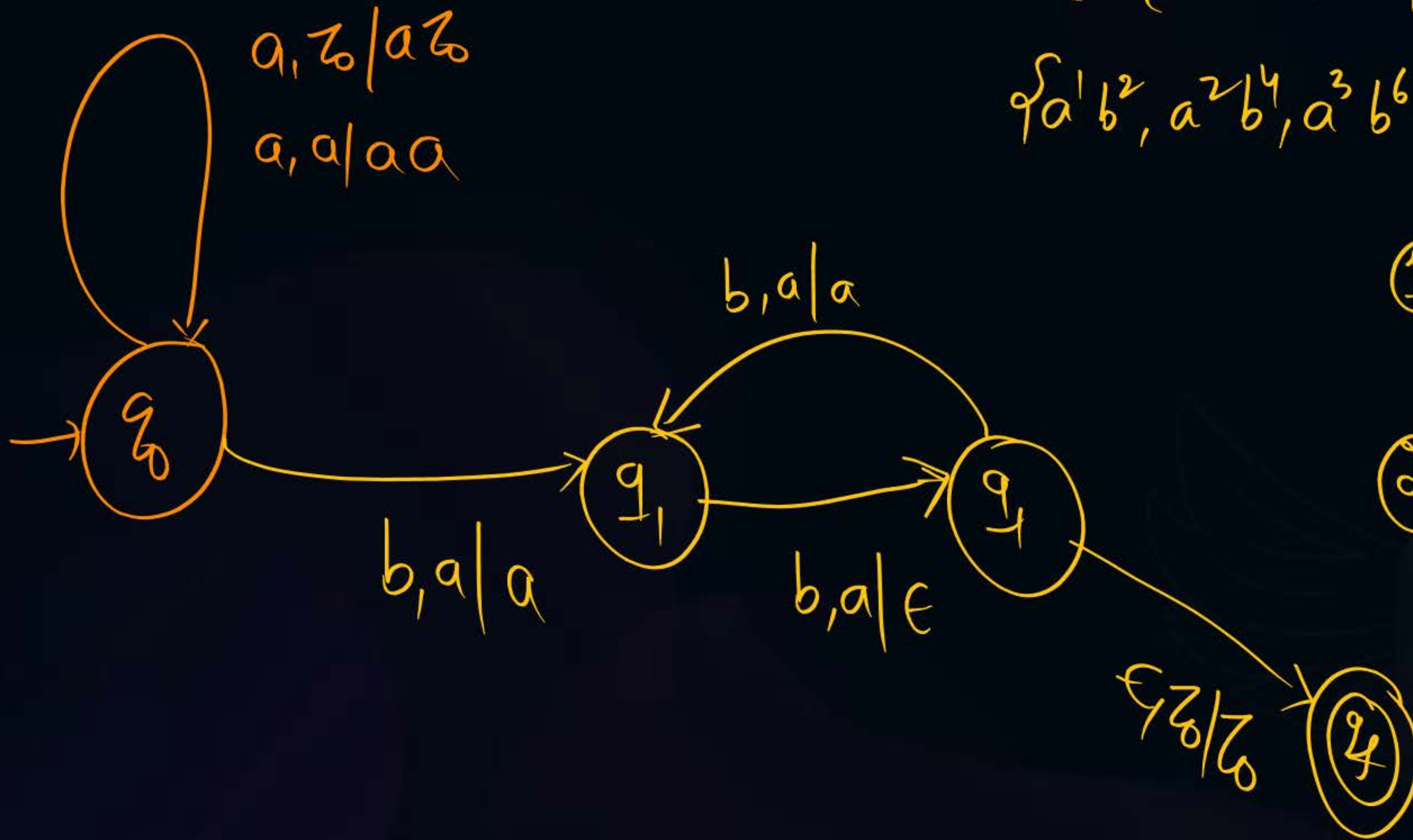


Topic : Pushdown Automata

Construct PDA for

$$L = \{a^n b^{2n} / n \geq 1\}$$

$\{a^1 b^2, a^2 b^4, a^3 b^6, \dots\}$ Logic



① push all a's into stack

②

1	$b \rightarrow \text{SKIP}$
	$b \rightarrow \text{POP}$



Topic : Pushdown Automata

Construct PDA for

$$L = \{ \boxed{a^n} \boxed{b^{2^n}} \mid n \geq 1 \}$$

PDA not possible Logic:

$$\{ \underline{(a^n)} \underline{(b^{2^n})} \}$$

PDA not possible

non CFL

$$L_1 = \{ a^n \underline{(b^{n^2})} \}$$

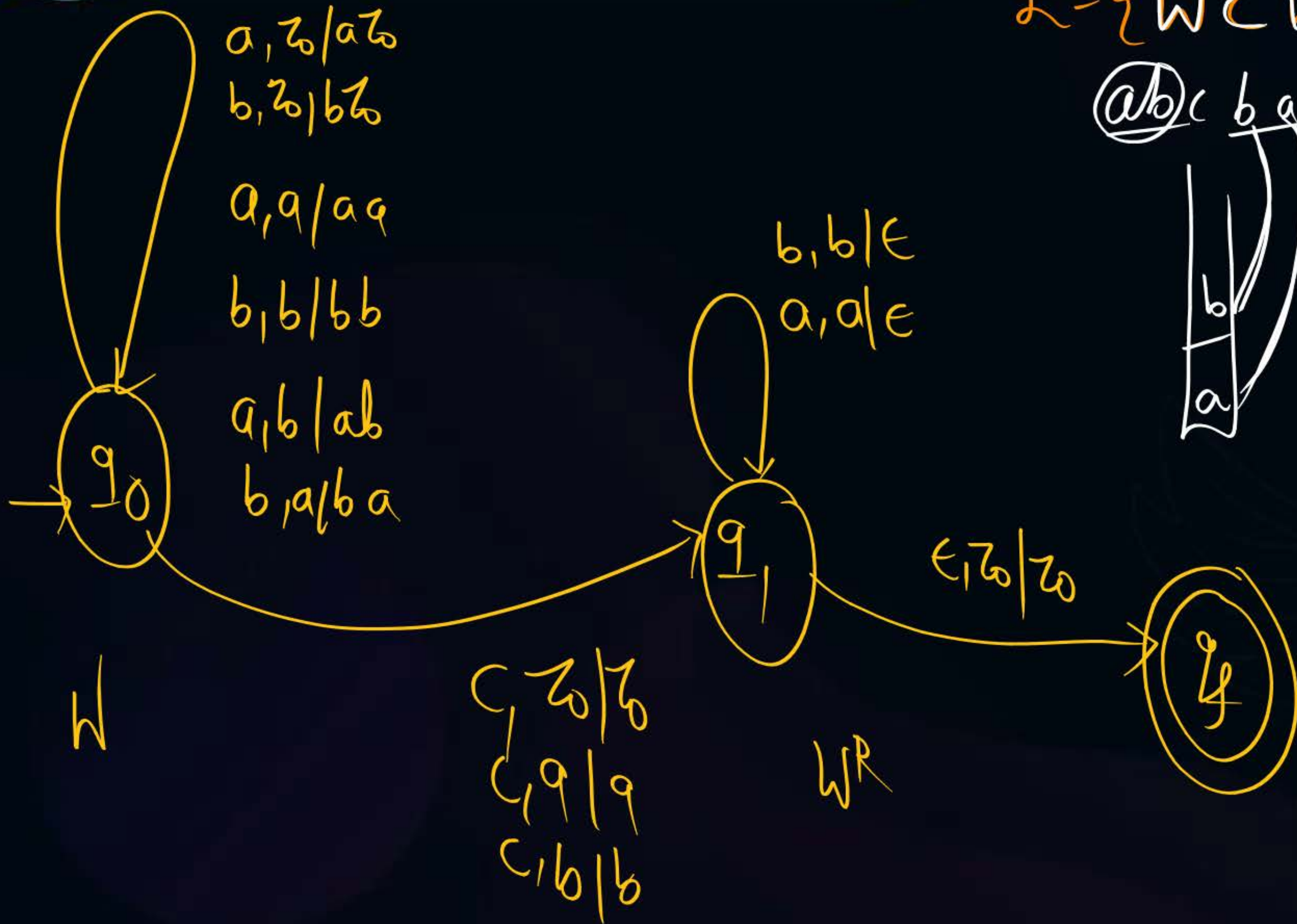
$$L_2 = \{ a^n \underline{(b^{n^n})} \}$$

$$L_3 = \{ a^p \underline{(b^p)} \mid p \text{ is prime number} \}$$



Topic : Pushdown Automata

Construct PDA for
 $L = \{W C W^R \mid W \in (a+b)^*\}$



- Logic
- ① $W \rightarrow \text{push}$
 - ② $C \rightarrow \text{SKIP}$
 - ③ $\begin{pmatrix} a, a \\ b, b \end{pmatrix} \rightarrow \text{pop}$



Topic : Pushdown Automata



Construct PDA for

$$L = \{ W^{\downarrow} W^R \mid W \in (a+b)^+ \}$$

W W^R abbbba

b
b
a
z

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

$$\delta(q_0, b, z_0) = (q_0, b z_0)$$

$$\delta(q_0, a, b) = (q_0, ab)$$

$$\delta(q_0, b, a) = (q_0, ba)$$

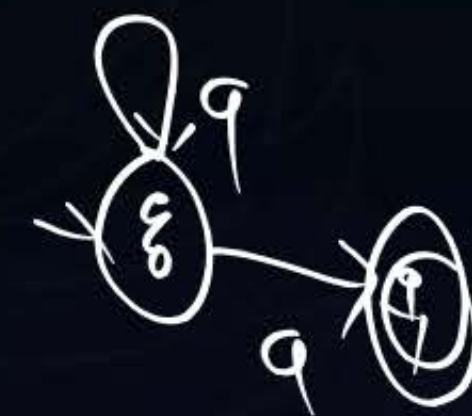
$$\delta(q_0, a, a) = (q_0, aa) (q_1, \epsilon)$$

$$\delta(q_0, b, b) = (q_0, bb) (q_1, \epsilon)$$

$$\delta(q_1, a, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, b) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_f, z_0)$$

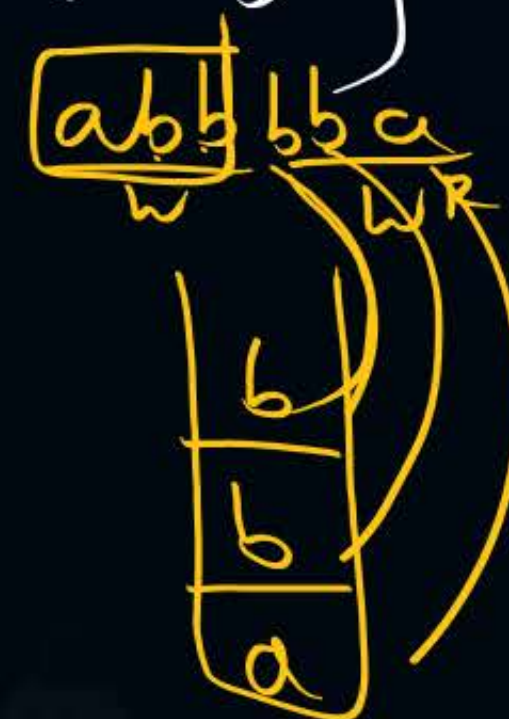




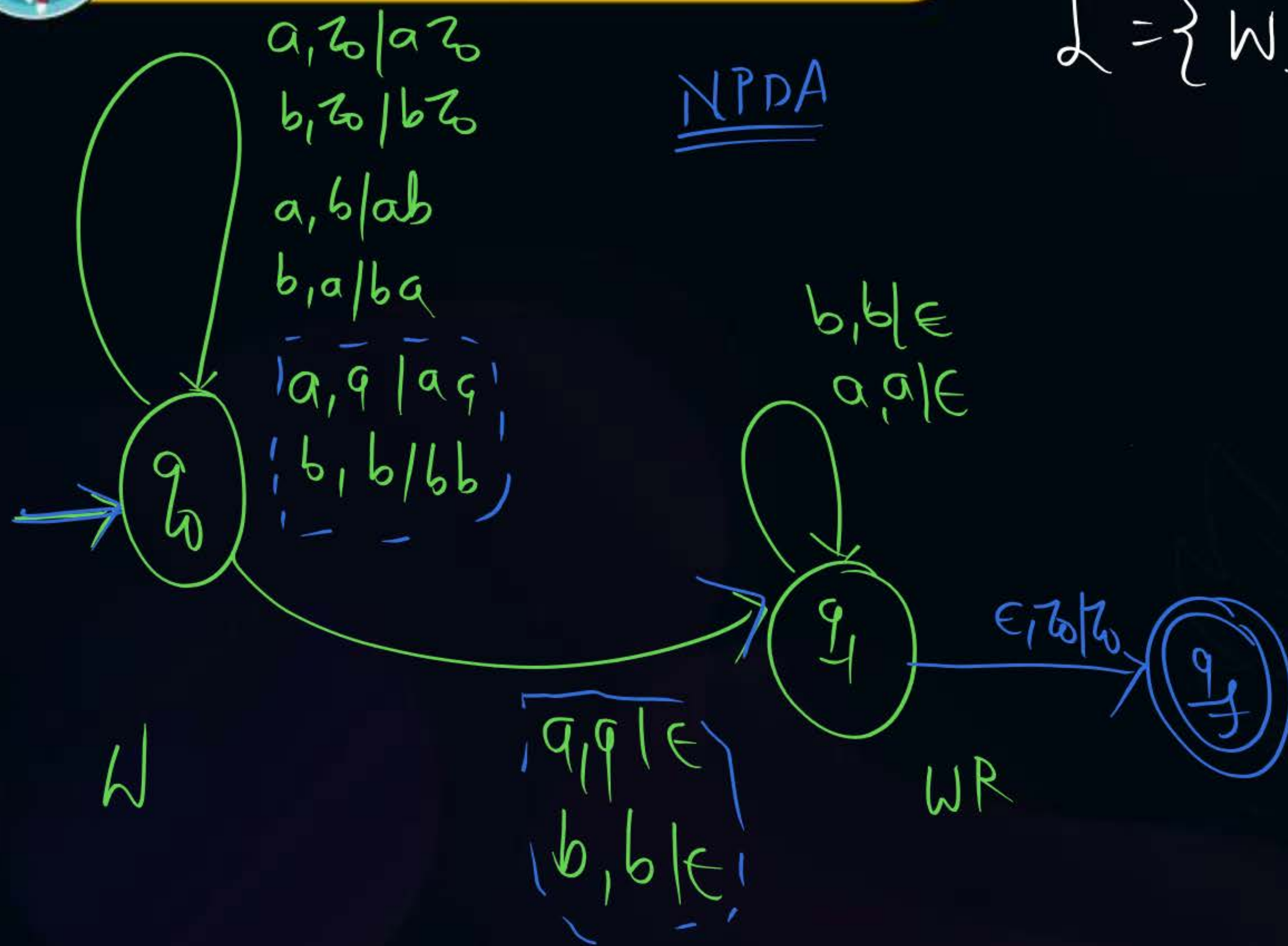
Topic : Pushdown Automata

Construct PDA for

$$L = \{ w^{\downarrow} w^R \mid w \in (a+b)^* \}$$



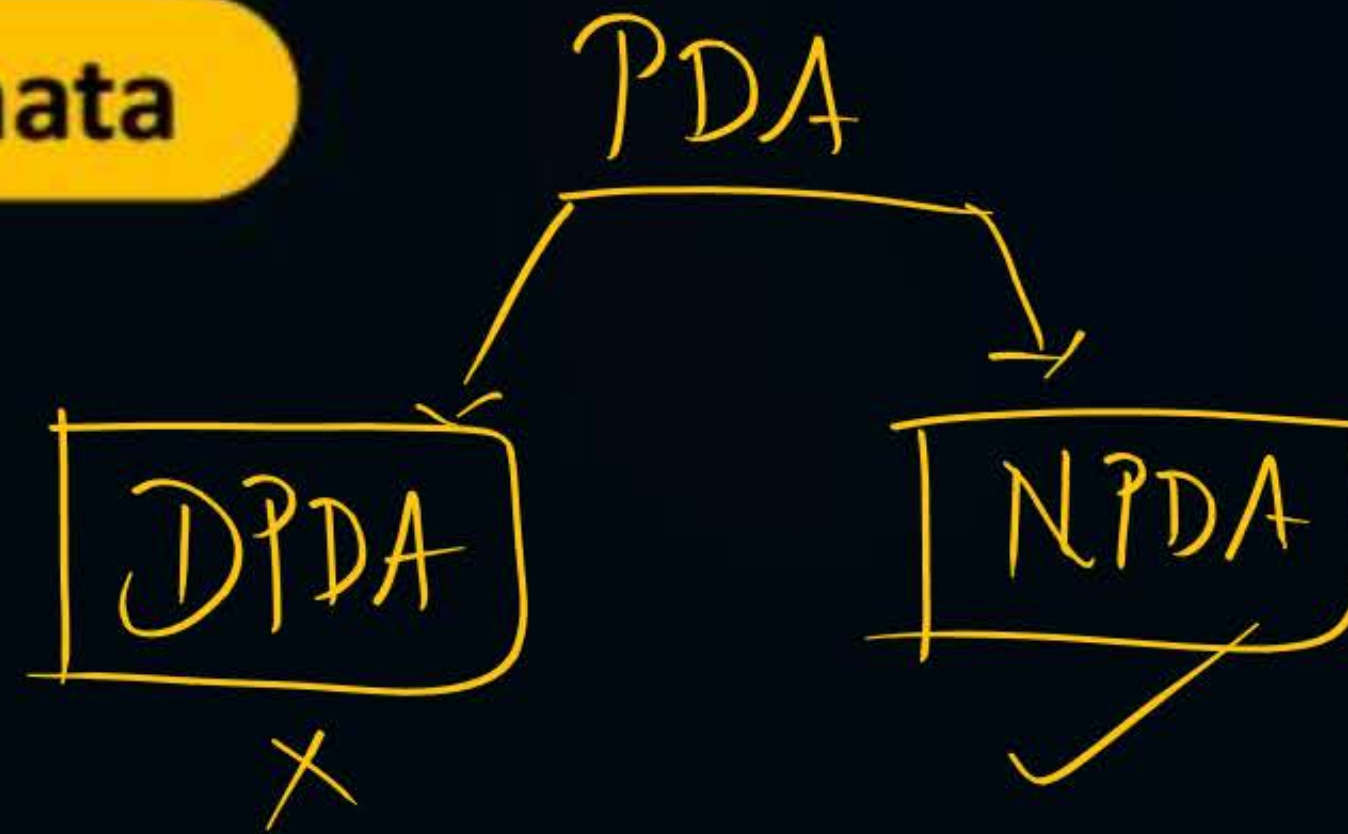
NPDA



$(q_0, b, b) =$



Topic : Pushdown Automata





2 mins Summary



Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU